MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS

NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 10 TIME: 11/11/2021

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We will discuss several questions from Chapter 8 in Ross' book.

Tutorial Questions

(1) Let $X_1, \dots, X_{20}$ be independent Poisson random variables with mean 1.
   (a) Use the Markov inequality to obtain a bound on
$$P\left\{\sum_{i=1}^{20} X_i > 15\right\}.$$
   (b) Use the central limit theorem to approximate the same probability in part (a).

(2) Let $X_1, \dots, X_n$ be independent Poisson random variables with mean 1. Let $X = \sum_{i=1}^{n} X_i$.
   (a) Use the Markov inequality to obtain a bound on
$$P\left\{\sum_{i=1}^{n} X_i > \frac{3}{2}n\right\}.$$
   (b) Use Chebyshev's inequality to obtain a bound on the same probability in part (a).
   (c) Derive a bound on the same probability in part (a) by considering
$$P\{|X - \mathbb{E}(X)| > t\} = P\{|X - \mathbb{E}(X)|^4 > t^4\} \leq \frac{\mathbb{E}(|X - \mathbb{E}\,X|^4)}{t^4}.$$
   You may want to use the results from Tutorial 4 Theoretical Exercise 19.
   (d) (optional) Derive a bound on the same probability in part (a) by considering
$$P\{X > t\} = P\{e^{\theta X} > e^{\theta t}\} \leq \frac{\mathbb{E}\, e^{\theta X}}{e^{\theta t}}, \quad \theta > 0$$
   and taking $\theta = \ln(t/n)$.

   You should see better and better upper bounds (better means smaller) from part (a) to part (d) for large n. This is a general phenomenon, not restricted to the sum of Poisson random variables. In general, in part (d), instead of taking θ to be $\ln(t/n)$ one can choose θ to minimize the upper bound $e^{-\theta t}\, \mathbb{E}\, e^{\theta X}$.

(3) Flip a fair coin 100 times. By using Central Limit Theorem, estimate the probability of more than 55 heads.

(4) If X is a gamma random variable with parameters $(n, 1)$, approximately how large need n be so that
$$P\left\{\left|\frac{X}{n} - 1\right| > 0.01\right\} \leq 0.01?$$

(5) Civil engineers believe that W, the amount of weight (in units of 1000 pounds) that a certain span of a bridge can withstand without structural damage resulting, is normally distributed with mean 400 and standard deviation 40. Suppose that the weight (again, in units of 1000 pounds) of a car is a random variable with mean 3 and standard deviation 0.3. Approximately how many cars would have to be on the bridge span for the probability of structural damage to exceed 0.1?

(6) A certain component is critical to the operation of an electrical system and must be replaced immediately upon failure. If the mean lifetime of this type of component is 100 hours and its standard deviation is 30 hours, how many of these components must be in stock so that the probability that the system is in continual operation for the next 2000 hours is at least 0.95?